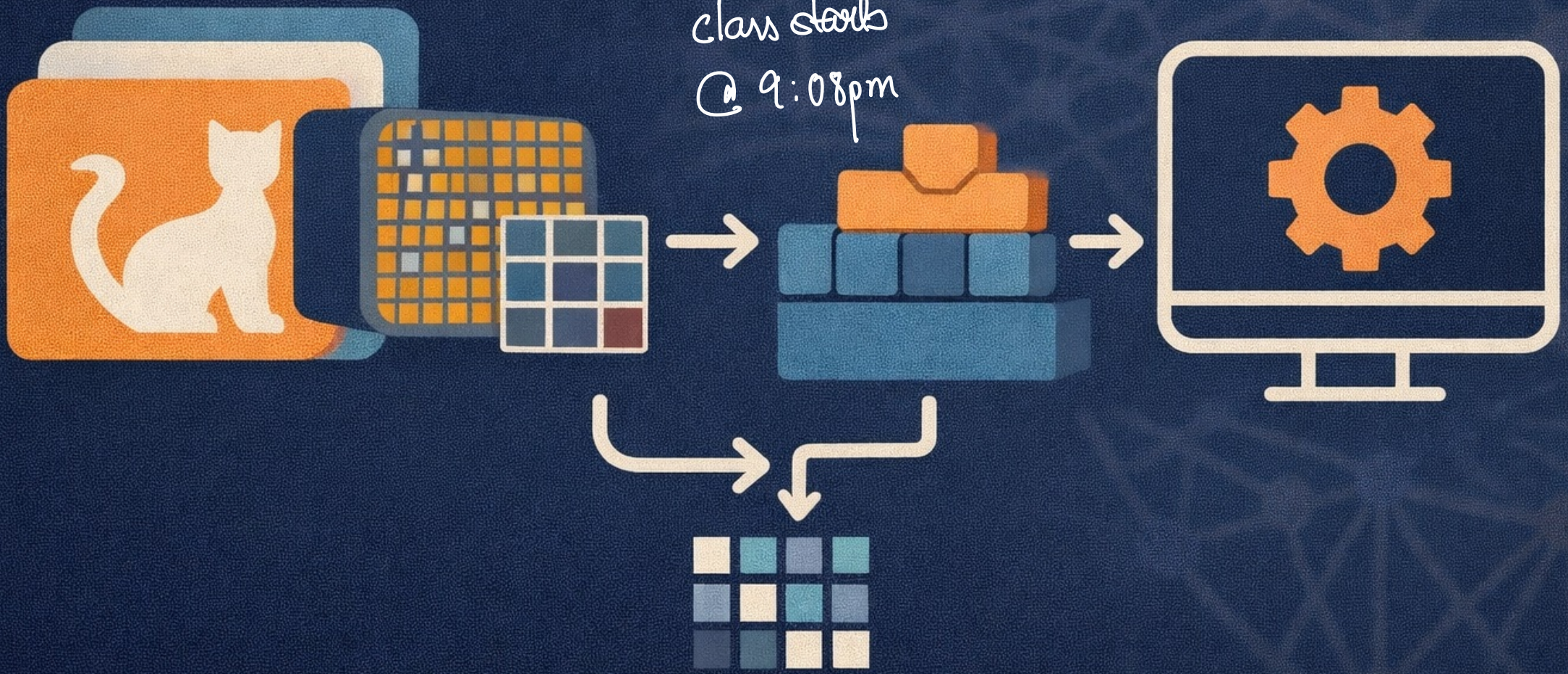
CONVOLUTIONAL NEURAL NETWORK



Lecture : Tackling Overfitting and Transfer Learning

#Agenda

1) Overfitting problems → Solution
2) Transfer learning.

How to fix overfitting problem?

1) Modification in Model pipeline (Architecture, loss fun^n, optimization)
2) Modification in Data pipeline (Data Augmentation)

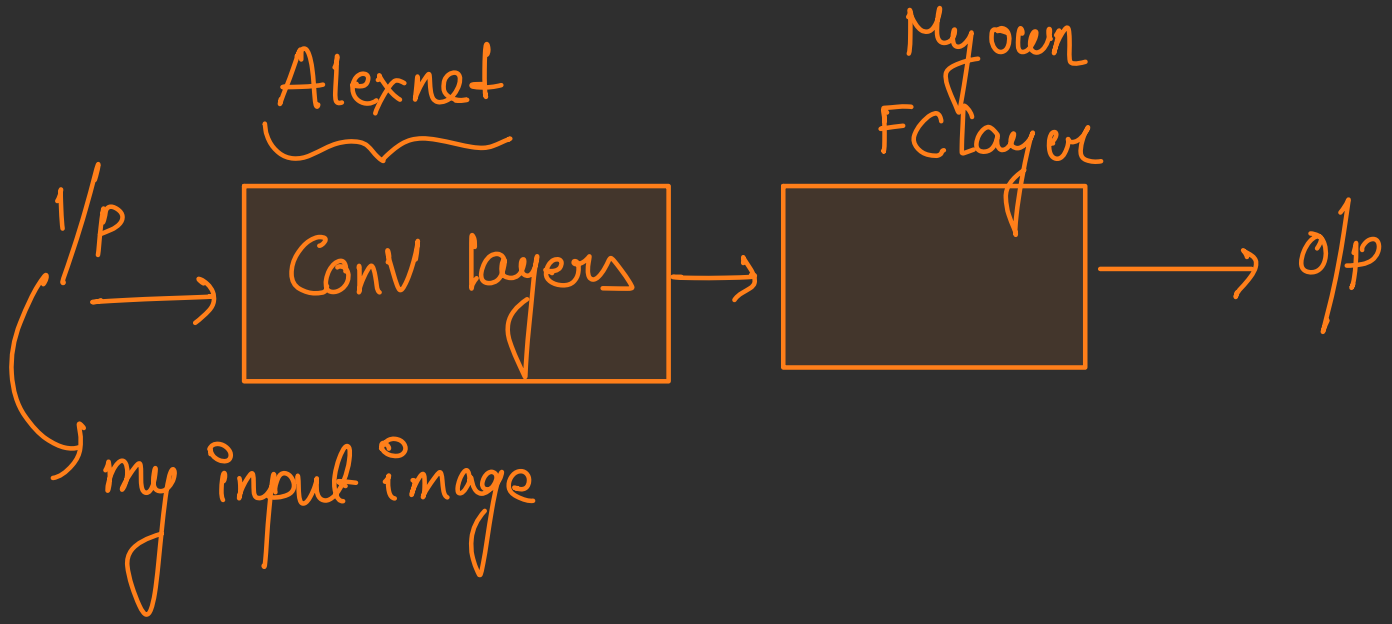
Summary

1) Tackle overfitting

- Modify Model pipelin
  1) Increase complexity
     - → pooling
     - → GAP
     - → Dropout & BN
     - → Early Stopping & Reduce LR Plateau
     - → L2 Regularization
- Modify data pipeline
  1) Data Augmentation
     - → Rotate
     - → Resize
     - → Flip
     - → Crop
     - → Brightness

2) Transfer Learning

↳ Use pre-trained model to improve accuracy for our problem.



100 x 5 Aug ⇒ 500

100 Input →

10 Epoch

ε_1 ⇒ 100

ε_2 : 100

ε_3 : 100

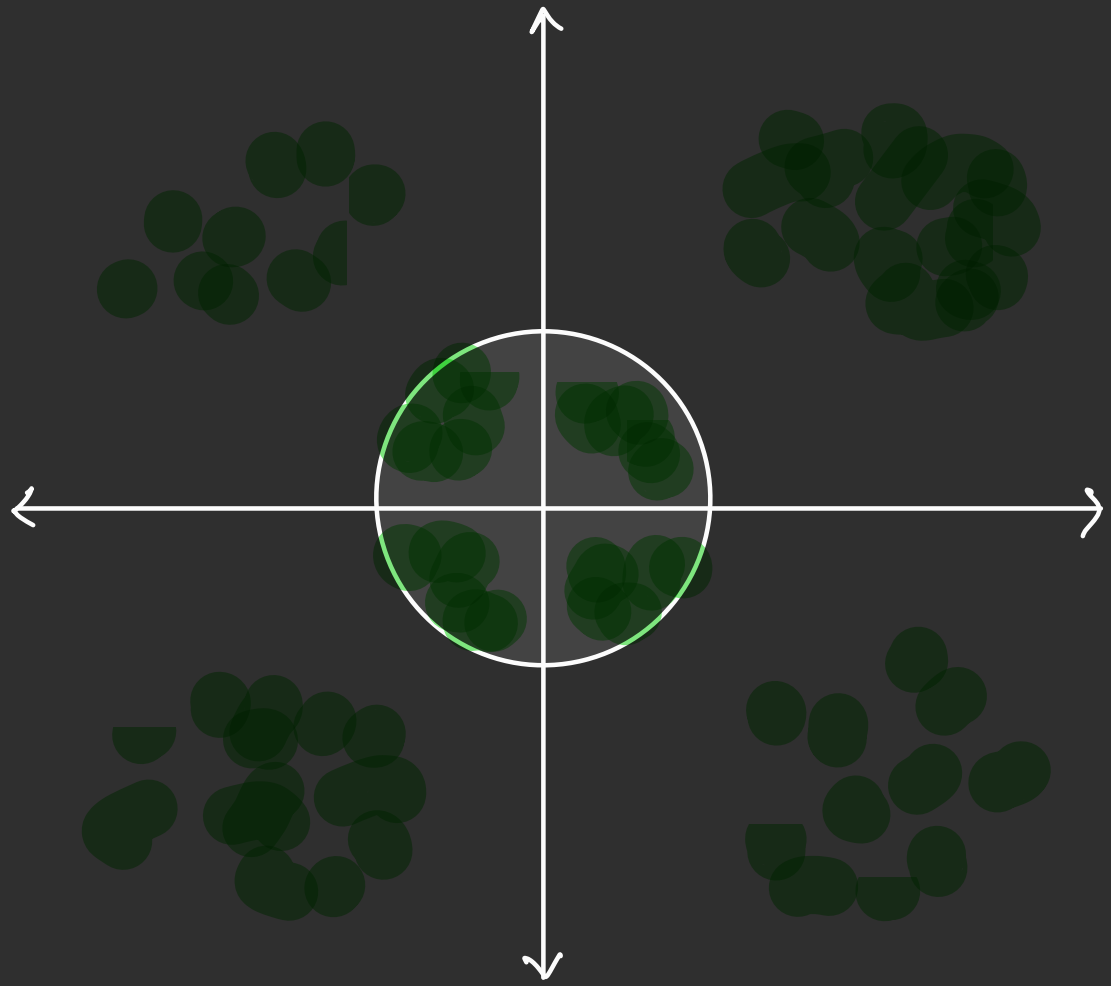
ε_4 : 100

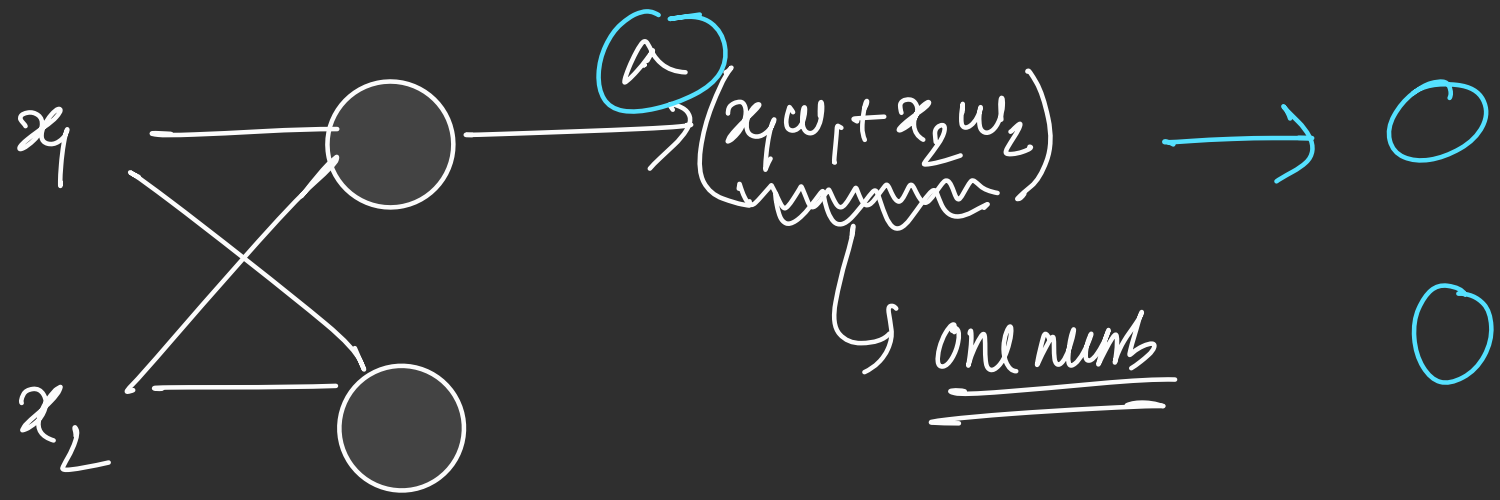
ε_5 : 100

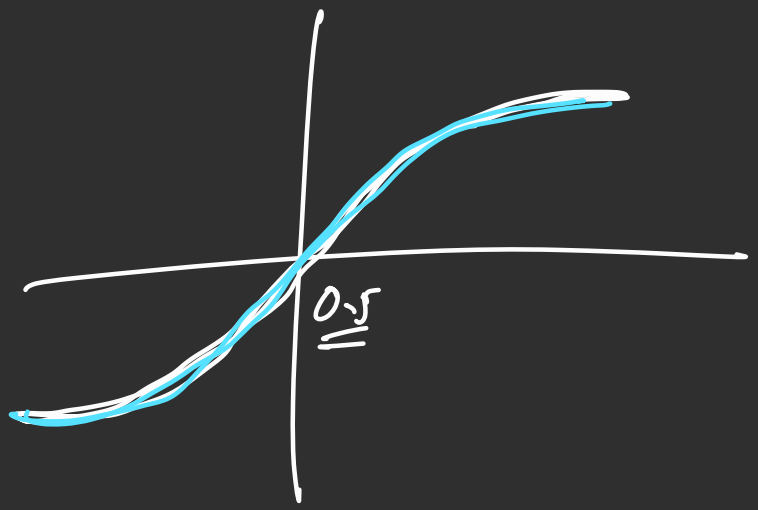


2012

NFT monkey







$$\sigma(z) = \frac{1}{(1 + e^{-z})}$$

$$0 = a(z)$$

$$z = w_1 x_1 + w_2 x_2 +$$

$$0_1 w_3 + 0_2 w_4$$

$$(w_1 x_1 + w_2 x_2) w_3 + (w_1 x_2 + w_2 x_1) w_4$$

$$w_1 x_1 w_3 + w_2 x_2 w_3 + w_1 x_2 w_4 + w_2 x_1 w_4$$

$$x_1 w_1 w_3 + x_1 w_2 w_4$$

$$x_1 (w_1 w_3 + w_2 w_4)$$

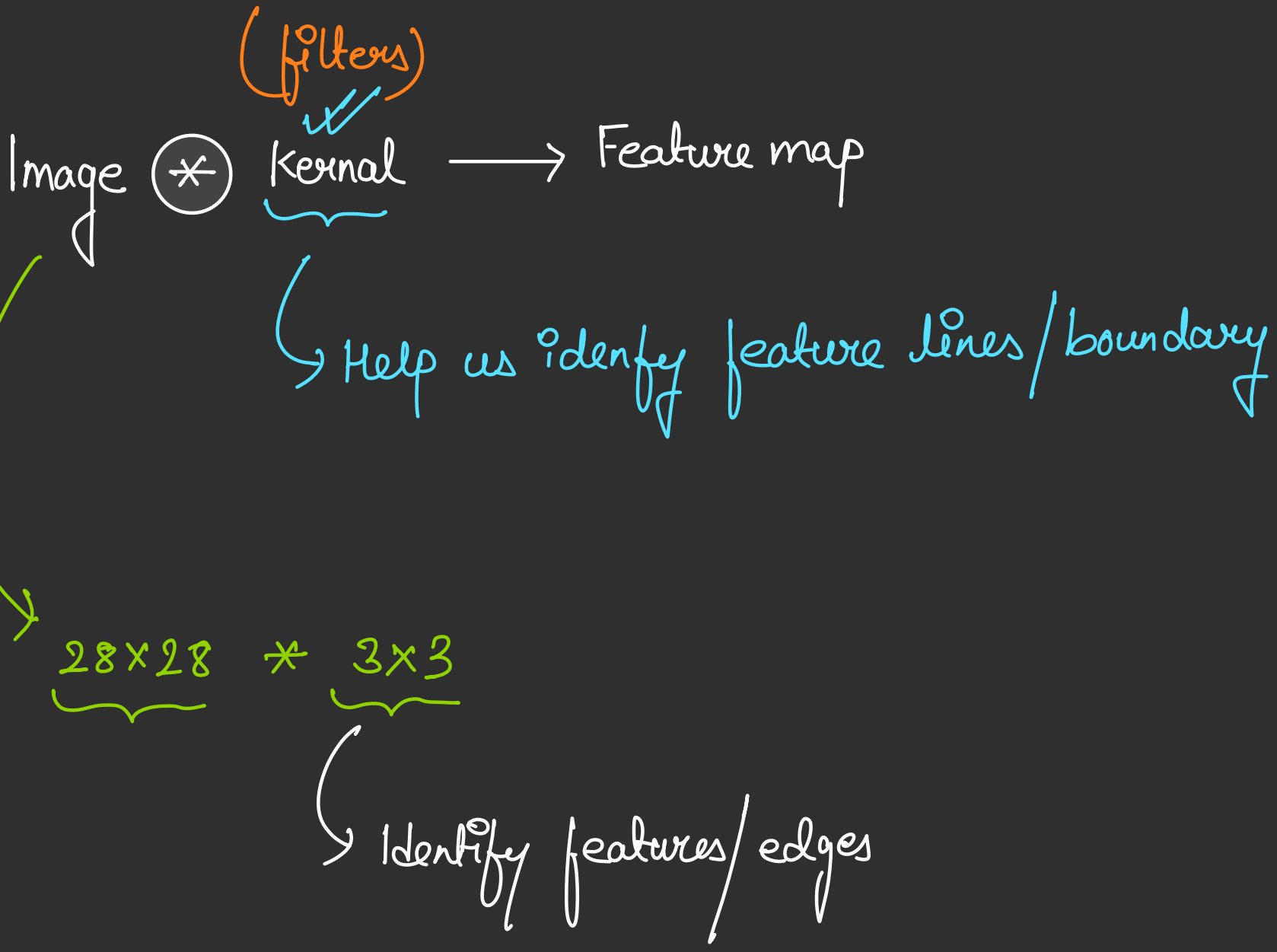
Recap

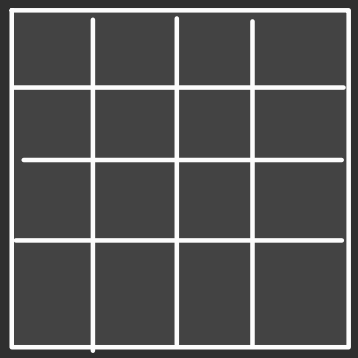
Image classification ⟶ ANN

28×28 ⟶ flatten (784) / feature ⟶ NN

28×28×3 (784 × 3)

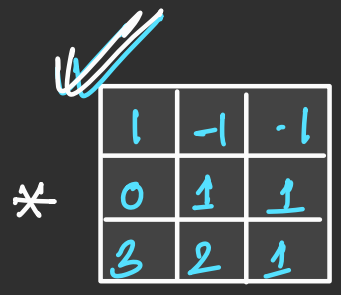
RGB


(filters)
Image * Kernal → Feature map
↳ Help us identify feature lines / boundary
28×28 * 3×3
↳ Identify features / edges



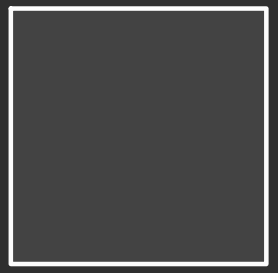
Image

(4x4)



Kernal

(3x3)



Feature map

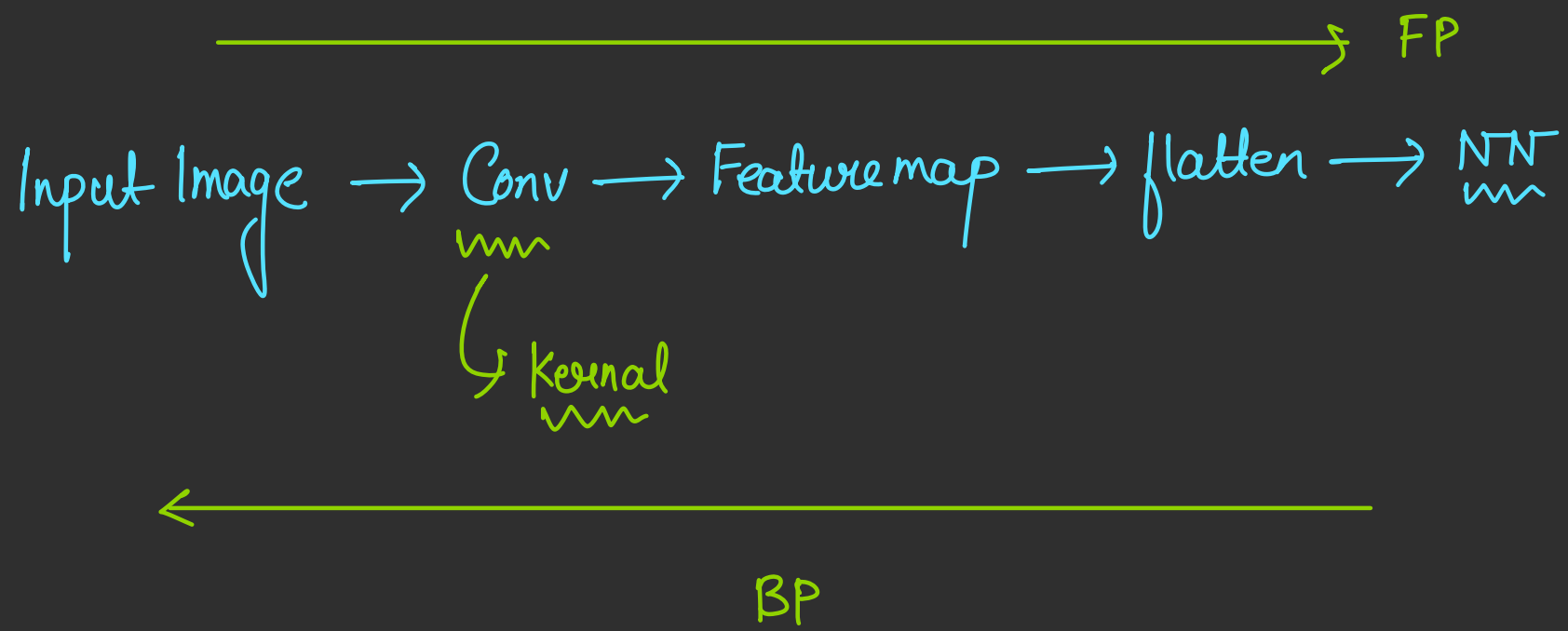
weights

RGB

4x4x3

Kernal

(3x3x3)


FP
Input Image
Conv
Feature map
flatten
NN
Kernal
BP

Output feature map size $= \left(\frac{n+2p-f}{s} + 1\right)$

- $n \Rightarrow$ size of input image
- $p \Rightarrow$ padding
- $s \Rightarrow$ stride
- $f \Rightarrow$ filter size

$\begin{cases} s = 1 \\ 2p \Rightarrow 0 \end{cases}$

Input = 5x5

filter $\Rightarrow$ 3x3

$\frac{5+0-3}{1} \Rightarrow 2+1 = 3$

3x3

output / feature map

Strides

np.arange(1, 10, 2)



$$\left(\frac{n + 2p - f}{s} + 1\right)$$

$s = 1$ $\Rightarrow$ $\left(\frac{5+0-3}{1} + 1\right) \Rightarrow 3 \times 3$

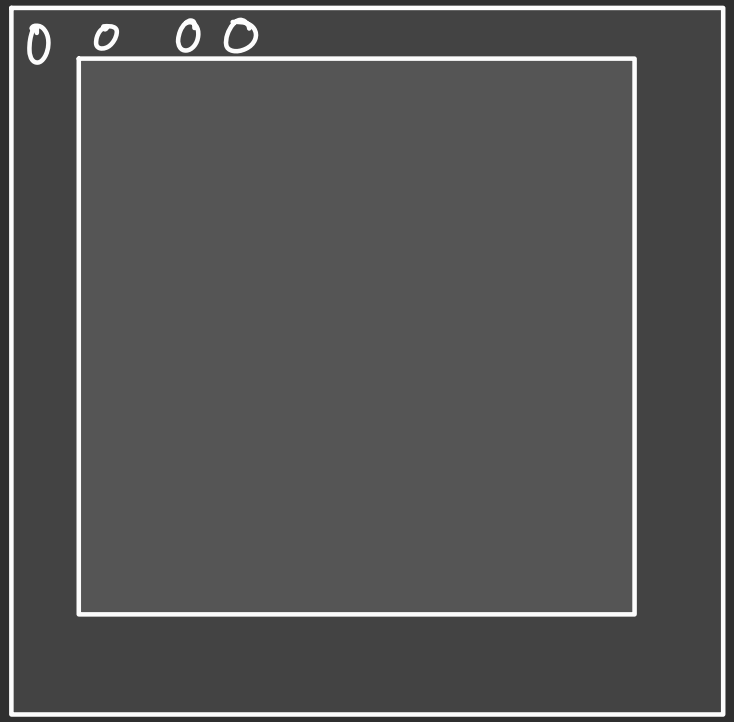
$s = 2$ $\left(\frac{5+0-3}{2} + 1\right) \Rightarrow 2 \times 2$

padding

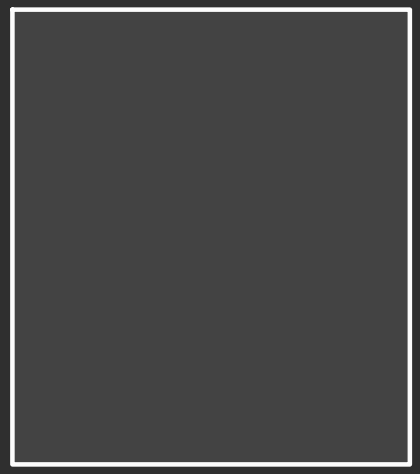
zero padding

same

valid



same Vs Valid



I/P

(28x28)

(3x3)

28x28

Feature map

26x26

Feature map

How to tackle an overfitting problem in CNN?

1) Modify your architecture / model pipeline.

2) Modify the data

"Data augmentation"

Summary

Input Image → ConvD → Feature map → Flatten → NN

Stride - You skip some information but helps to train faster

Skipping pixel in images

padding → same (Padding zero)

valid (No padding)

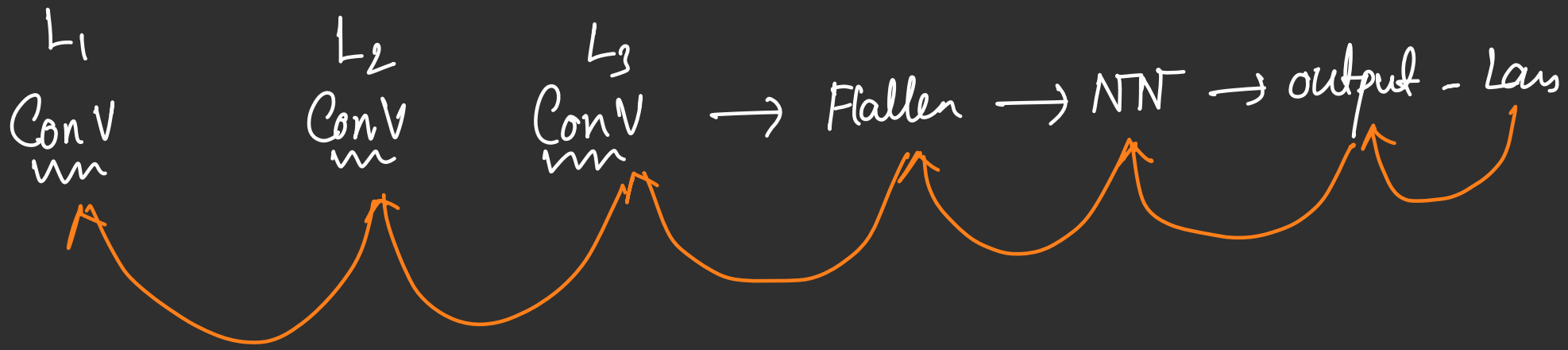
$$\text{Feature map size} = \left(\frac{n + 2p - f}{s} + 1 \right)$$

pooling ⟶ To reduce feature map size, Focus more on important aspect

↳ Max pooling, Avg pooling

Overfitting

1) Modify the architecture
2) Modify the data set



{ YOLO
Resnet 50

B G P R

B R G P